

Tobor

A service that turns a broken part, an unfinished idea, or a supported robot fault into a verified, working solution.

Prepared for Prakash. This plan converts the supplied tweet into an India-first business using your existing 10+ 3D printers and IoT equipment. Tobor is a working name drawn from this project context; naming, trademark and domain availability have not been checked.

The product	The factory	The business
One portal for custom parts, replacement parts and supported robot repairs.	An owned engineering, printing, electronics and inspection workshop, supported by qualified manufacturing partners.	Earn from diagnosis, design, manufacturing, repair and repeat supply; add account retainers after repeat demand is demonstrated.

Recommendation: launch as a managed service in one city, led by two customer groups: small factories/assembly teams and robotics/IoT labs. Accept selected consumer replacement jobs through repair-shop partners. Expand the catalog as each part family and repair procedure becomes repeatable.

What your existing setup changes: it reduces the need for printing hardware and prototype electronics. The next investment should address measurement, engineering, reliable production, testing and customer acquisition. Ten printers alone do not establish qualified manufacturing capacity.

Planning assumptions: India, INR, approximately 1,200-1,500 sq ft premises, four operating staff plus founder oversight, and existing software-team capacity for a lean portal. Location, printer models, material capability, existing tools and budget remain unverified.

Evidence boundary: the tweet text is supplied by you and is used as inspiration. Its claims about a specific model or a new manufacturing era are not prerequisites for this business. This plan assumes AI can assist drafting; physical correctness must be established by engineering and tests.

Read this PDF alongside **Tobor_Factory_BOM_and_Financial_Model.xlsx**. The workbook contains item-level quantities, planning cost ranges, inventory deductions, monthly costs and editable financial/capacity assumptions. This is a plan for review; no equipment has been ordered and no portal has been deployed.

01 / The product promise

Customer promise: "Show us what is broken or what you need. We will assess it, propose a solution, and source, make or repair it with agreed checks." A customer should not need to know CAD, printing processes or electronic part numbers to begin.

Service	Typical request	Deliverable
Make a part	I need a sensor mount, enclosure or assembly fixture.	Approved design, manufactured part, inspection record and reorder ID.
Replace a broken part	This knob or housing is broken and the spare is unavailable.	Failure assessment, sourced or engineered replacement, fit confirmation.
Fix a robot/device	My supported lab robot has a damaged body or intermittent sensor.	Diagnostic report, approved repair, configuration record and functional test.
Repeat a verified part	Make another batch of the part you supplied last month.	Same approved revision, traceable process and batch inspection.

One service decision: source an existing spare, repair the component, manufacture a replacement, use a qualified partner, or decline. If an original spare is cheaper and more reliable, offer it. The factory exists to fulfill the best route, rather than forcing every job through a printer.

Who owns delivery: Tobor is the contracting service provider for its accepted work. A case owner coordinates design, factory work, customer updates and quality checks. Partner referrals must state whether Tobor remains responsible for fulfillment or the customer contracts directly with the specialist.

What "any object" means: a broad intake interface with a clear feasibility assessment. It cannot mean a guarantee to reproduce every material, repair every robot or safely substitute every component. Show supported categories and limitations before payment.

Long-term product: a service history and spare-parts system for physical assets. Each successfully installed design becomes a private, versioned part record. Repeat orders can bypass unnecessary redesign while retaining inspection and change control.

The durable advantage is accumulated evidence: which design fits which asset, under which conditions, made by which process, and with what repair outcome.

02 / Where to start

Priority	Customer and first offers	Why / boundary
1	Small factories, electronics assembly teams and integrators: assembly nests, inspection holders, sensor mounts, cable guides, noncritical knobs.	Clear interfaces and repeat use. Begin with low-risk tools/peripherals; avoid machine safety and high-force tooling.
2	Robotics startups, education labs and IoT builders: enclosures, brackets, harnesses, replaceable sensors/modules and supported repairs.	Uses both printers and electronics capability. Publish a supported equipment list.
Channel pilot	Independent repair shops: missing clips, feet, covers, spacers and knobs.	Shops consolidate consumer demand and can check fit. Batch small orders to control shipping/design cost.
Later	Facilities, gyms, hospitality and clinics: external accessories, organizers and cosmetic parts.	Only suitable noncritical applications; not loaded gym components, food-contact parts or patient-contact devices.

Your manufacturing/MES and IoT experience makes industrial peripherals and assembly tooling a credible first wedge. Existing relationships can provide discovery access, but they are not assumed customers or commitments. Ask for real failed parts and purchase histories before creating a sales forecast.

Initial catalog: 12 demonstrators. Three assembly/inspection fixtures, three sensor/mounting solutions, two IoT enclosures, two noncritical replacement parts and two supported robot repair demonstrations. Include before/after photos, accepted dimensions, actual lead time and the test performed. Obtain permission before publishing customer work.

Local launch: operate in the city where the printers and team already sit. If that is Chennai, begin with reachable industrial areas and labs; do not move the factory based on this assumption. Use local pickup for fit-sensitive jobs and courier service for standardized parts.

Demand evidence to collect: frequency of failure, current workaround, current spare cost/lead time, downtime or manual effort, approver, budget, installation responsibility and probable reorder interval. Separate urgent operational demand from hobby curiosity.

Vendor application evidence supports the feasibility of printed manufacturing aids, including assembly and inspection fixtures. It does not establish Tobor demand or the savings a specific buyer will achieve. [S1, S2]

03 / Market position

Provider	Published capability	Tobor implication
Fracktal Works	Sketch-to-CAD engineering, additive manufacturing, CNC and sheet metal. [S3]	General design/manufacturing already has capable competitors.
3Ding	Printing, scanning, modeling and advertised AI quotation. [S4]	A generic upload-and-print page is insufficient differentiation.
Xometry	Design marketplace including reverse engineering and manufacturing workflows. [S5]	Integrated local diagnosis and installed-fit support should be tested as the distinction.
ABB / specialist repair firms	Robot workshop/field repair, spares and maintenance services. [S6]	Industrial robotics needs specialist capability and partnerships.
Philips Fixables	OEM-published printable replacement parts and upgrades. [S7]	Authorized digital-spares fulfillment is a plausible later channel.

Positioning to validate: "One accountable service for the part and the repair." Combine issue-led intake, source-or-make choice, measured fit, electronics diagnosis, installation feedback and repeat supply. Do not claim these capabilities are unique; measure whether the combination solves a local service gap.

Revenue progression: paid engineering and repair first; repeat manufacture second; supported asset accounts and authorized catalogs third. Platform commissions or software licensing belong after the internal operation is demonstrably repeatable.

Market sizing: use a bottom-up model. For example, 30 paying business accounts x 2 orders/month x INR 6,000 = INR 3.6 lakh/month; 100 suitable consumer/repair-shop orders x INR 2,000 = INR 2 lakh/month. These are hypothetical account counts and basket sizes, not a forecast or TAM estimate. Track the funnels separately to avoid counting a partner order twice.

Policy context: India has a Right to Repair portal that provides repair-related information. This is useful ecosystem context, not blanket permission to copy proprietary designs, use trademarks or disregard product-specific requirements. [S8]

Win a narrow set of repeatable jobs before promising a national repair marketplace. Competing on filament price would underuse your engineering and IoT advantage.

04 / Service scope and qualification

Route	Launch examples	Required condition
In-house	Knobs, cosmetic covers, feet, spacers, cable guides, low-load mounts and simple enclosures.	Documented use, qualified material/process and measurable fit.
In-house, engineer-led	Assembly nests, inspection holders, supported robot housings, 5-24 V bench IoT modules and harness work.	Trained staff, documented procedure and controlled functional test. The voltage range is a proposed service limit, not a universal safety classification.
Qualified partner	CNC metal, SLS/MJF nylon, sheet metal, specialist coatings, PCB manufacture/assembly, complex scanning/metrology.	Written specification, traceable supplier, first-article inspection and agreed acceptance.
Specialist track	Industrial robot joints, brakes, safety controllers, proprietary calibration, mains power repair, high-energy batteries.	Named competent provider/OEM route; explicit responsibility and recommissioning plan.
Outside launch scope	Brakes/steering, lifting/load-bearing safety parts, pressure/gas components, medical guides, food-contact parts, safety interlock modifications.	Do not accept production under the launch capability. Assess only for referral.

The consequence of failure determines the route. A cosmetic cap on gym equipment can be suitable; a load-bearing pulley on the same machine may not be. A gripper finger is not automatically low-risk: load, speed, dropped-object consequence and access to people must be assessed.

Feasibility form: intended use; what failed and why; load and duty cycle; temperature, sunlight, liquid and chemical exposure; electrical or pressure exposure; required life; mating surfaces; accuracy; applicable customer/OEM rules; consequence of failure; and who installs/tests the part.

Approval ownership: the engineering lead accepts the technical scope. Customer approval confirms scope, appearance, interfaces, price and delivery expectations; it does not certify safety or remove Tobor engineering responsibility.

Universal Robots specifies trained repair personnel and service-manual procedures, including post-repair validation. This illustrates why "robot repair" must be scoped by equipment family and technician competence. [S9]

05 / Customer journeys

Journey	Steps	Customer receives
Missing/broken part	Upload photos and dimensions -> assessment -> send sample/mating part if needed -> design quote -> prototype/fit -> production -> delivery.	Clear measurements needed, approved drawing, part, fit instructions and reorder record.
New custom part	Describe function or upload CAD -> requirements/DFM -> engineering quote -> design approval -> first article -> batch.	Editable design deliverables if purchased, prototype evidence, quantity pricing and batch report.
Robot/device repair	Identify model and symptoms -> remote triage -> receive/record asset -> paid diagnosis -> repair estimate -> authorized work -> function test -> return.	Diagnostic findings, replaced-part list, configuration changes, test report and scoped warranty.

Capture instructions: photograph the whole object, the failed part in place, every accessible side, broken faces and the mating interface. Include a ruler/reference and state units. Ask for brand/model/serial, quantity, urgency and original spare availability. For hidden geometry or tight fit, arrange a physical sample rather than guessing.

Quote in stages: show a preliminary range only when information is incomplete. Sell a defined diagnosis or design task before promising a final repair/manufacturing price. After assessment, provide itemized engineering, parts/process, finishing, testing, logistics, taxes and customer approval conditions.

Timeline targets for the pilot: acknowledge within one business day; assess a complete simple request within two; target simple approved printed parts in 3-5 business days and reverse-engineered prototypes in 5-10. Repair lead time starts after diagnosis, approval and spare availability. These are internal planning targets, not launch guarantees.

Exceptions visible to customers: need dimensions/sample, awaiting spare, changed scope, customer decision, rework, partner referral and unable to fulfill. Do not leave a job in "processing" while its delivery clock silently slips.

Economically unsuitable jobs: explain when design plus shipping exceeds replacement value. Offer a shop batch, an existing spare or return of the item with the agreed assessment fee. Record the reason so the catalog and marketing improve.

06 / Portal requirements

Module	Minimum behavior	Acceptance evidence
Guided intake	Three request types; photos/video/files; model/serial; dimensions and units; use/risk questions; sample option.	Engineer can identify missing critical information without an unstructured chat search.
Cases and assets	One case owner; parent robot/asset with linked repair/part work orders; status timeline and messages.	Every task, file, quote and shipment resolves to a case/asset.
Assessment and quote	Source/make/repair/partner decision; itemized versioned quote; tax/shipping fields; expiry; deposit tracking.	Customer approves the exact scope and price version.
Design review	3D preview, dimensioned drawing, unknown dimensions, revision comparison and approvals.	Critical unknowns block release; approvals bind to exact revision.
Factory job card	Approved part/revision, quantity, machine/material/profile, tasks, due date and traveler QR.	Operator receives only the released manufacturing package.
Quality and repair report	Measured dimensions, pass/fail criteria, test photos/video, fault/action list and release identity.	Failed or missing required checks block dispatch.
Payments and logistics	Gateway-hosted payments, invoice, pickup/sample tracking, shipment and balance/refund record.	Payment/dispatch state reconciles to the case, including partial/cancelled work.
Private part library	Asset-linked approved part, permissions, revision history and reorder request.	Repeat request confirms unchanged interface/use and allowed design rights.
Operations dashboard	Backlog, aging, engineer/machine load, rework, contribution and supplier delays.	Team can see bottlenecks and actual versus quoted effort.

Roles: customer, organization buyer, case owner, design/repair engineer, operator, QC/release lead, administrator and limited external partner. In a small team one person may hold more than one role; record who performs each approval. Require an independent second review for higher-consequence accepted jobs.

Defer: open supplier marketplace, fully automatic CAD-to-machine execution, native mobile apps, broad field-service dispatch, predictive failure AI and public design marketplace. Start with a responsive web portal; a staff-assisted phone/WhatsApp channel can feed the same case record.

07 / Data and software architecture

Recommended implementation shape: a modular application with a web frontend, API, relational database, private file storage and asynchronous worker queue. Start with one deployable backend and separate resource-limited geometry workers; do not begin with a large microservice estate.

Layer	Proposed choice / role
Customer/operations UI	React or Next.js frontend; request forms, quotes, model preview, approval and factory queue.
Business API	NestJS/TypeScript or the team's established stack; case state machine, RBAC, pricing and job orchestration.
System of record	PostgreSQL for customers, assets, cases, specifications, revisions, approvals, jobs and quality records.
Files and background work	Private object storage with versioning; queued scanning/conversion/thumbnails/CAD jobs; backups and restore checks.
Engineering worker	Python + CadQuery/FreeCAD for selected parametric templates, CAD export and geometric validation; commercial CAD when required.
Workshop gateway	Read-only telemetry collection first; supported printer adapters selected after hardware audit; staff authorizes production.
Integrations	Payment provider, courier/tracking, accounting export and notification service; adapters with retries and reconciliation.

CAD record: keep native editable source and STEP plus a dimensioned drawing. STL/3MF and viewer meshes are derived outputs. CadQuery supports script-based parametric models and standard CAD exports; it is a practical tool to evaluate for constrained part families. [S10]

Revision chain: specification v1 -> CAD/drawing v1 -> engineering release -> customer approval -> manufacturing package -> first article -> batch release. Bind approval to document IDs/hashes. A change creates v2 and invalidates affected downstream approvals. Never overwrite an approved design.

Security: private designs by default, organization-level authorization, expiring download links, encrypted storage, audit events, malware/file checks and tested backups. Isolate customer CAD/scripts and uploaded files with CPU/memory/time limits and restricted network access. Never treat uploaded G-code as an approved machine job.

IP and repair data: customer rights to supplied designs, license for manufacture and optional design ownership/delivery must be explicit. Obtain consent for template reuse, model-provider processing, factory reset and configuration changes. Do not automatically add customer files or robot logs to training data or a shared catalog.

08 / AI-assisted engineering

AI can assist	Required control
Classify a request and ask missing questions	Engineer verifies intended use and consequences of failure.
Draft a parametric model from a validated template	Dimensions are tagged measured, customer-provided, inferred or unknown. Critical unknown/inferred values block production.
Suggest material/process from an approved catalog	Engineer checks load, environment, orientation, life and qualified capability.
Summarize manuals and diagnostic possibilities	Link the exact model/manual source; technician verifies before touching equipment.
Draft quotations and test checklists	Use observed costs and approved procedures; humans approve scope and price.
Flag print anomalies in images or telemetry	Alerts request review; they are not proof of quality or the sole safety control.

Photo-to-CAD reality: an image cannot prove hidden dimensions, material grade, internal damage or fit. Photogrammetry requires a scale reference and may produce visually convincing but dimensionally unreliable geometry. Scanning does not eliminate metrology of critical interfaces. [S11]

Initial AI use cases: classify the 12 demonstrators; prepare structured requirements; draft three to five constrained parametric families such as brackets, spacers, covers and enclosures; produce inspection-sheet drafts; and summarize customer-approved repair reports. Human CAD remains a normal route.

Evaluation set: 20 previously measured, low-risk jobs with reference drawings. Compare manual and assisted workflows for total engineering minutes, dimension errors, invalid solids, first-fit acceptance and rework. Measure end-to-end time including correction, not just generation speed. Only scale a template after repeated physical success.

Non-negotiable machine boundary: the model cannot release dimensions, approve a safety-related design, change robot safety settings, flash arbitrary firmware or start machinery. A qualified person selects the released revision, compatible machine/material/profile and inspection plan.

Model flexibility: select a model using these real-job evaluations, cost, privacy terms and tool reliability. Do not make the business dependent on the tweet's model claim. Geometry, job history and acceptance criteria must remain portable between AI providers.

09 / Use the equipment you already own

Existing asset	Best first use	Audit before relying on it
10+ 3D printers	Short-run parts, prototypes, fixtures, enclosures and repeat batches.	Exact model/process, build volume, enclosure, nozzle, temperatures, materials, thermal protection, maintenance, networking and repeatability.
IoT development boards	Test fixtures, protocol simulators, telemetry gateways and prototype controllers.	Board revision, electrical ratings, condition, firmware/license and available quantities.
Sensors and actuators	Controlled robot/sensor diagnosis; motor/encoder test rigs; fixture feedback.	Measurement reliability, compatible voltages, connectors and traceable replacements.
Connectivity kits	Test serial, I2C, SPI, CAN and RS-485 where supported; collect machine status.	Adapters, isolation where appropriate, protocol documentation and trained operators.
Existing PCs/network/tools	CAD, job cards, documentation and local gateway.	Commercial software rights, performance, security, backups and tool calibration.

Suggested allocation if all ten qualify: six for repeat production, two for prototypes/new materials, one for longer jobs and one available for maintenance/overflow. This is a scheduling starting point; route by actual material and process capability. Do not reserve incompatible printers for an engineering material merely to fill the allocation.

Qualification: inspect each machine; service it; calibrate; print dimensional and fit coupons; test representative parts; record measured limits. Qualify the printer-material-nozzle-profile combination rather than the printer name. Establish recheck triggers after maintenance or a process change. There is no universal FDM fit tolerance; geometry, orientation and material all matter. [S12]

IoT as internal product development: build a simple uptime/material-dryness dashboard, fixture cycle counter and instrumented test harness. Use these to gather evidence and improve service. Certified detection, electrical protection and emergency isolation must operate independently of prototype monitoring.

Production components: unknown hobby-kit stock is useful for learning and prototyping. Customer builds need specified, rated, traceable parts with reproducible supply. Reuse only inspected equipment; create a separate stock ledger for purchased production components.

10 / Factory layout and utilities

Zone	Indicative area	Purpose
Receiving and quarantine	100 sq ft	Record incoming asset condition, serial, accessories and sample chain of custody.
CAD and clean metrology	150 sq ft	Design reviews, critical measurement and document control.
Print cell	300 sq ft	Ten-machine layout, access, maintenance clearance, enclosure/extraction and dry storage access.
Electronics benches	150 sq ft	ESD work, soldering extraction and low-voltage diagnosis.
Assembly and guarded test	180 sq ft	Fixtures, controlled robot motion, assembly and final functional checks.
Finishing/dirty work	100 sq ft	Sanding, drilling and dust control separated from electronics/metrology.
Materials, packing and finished goods	140 sq ft	Dry filament, traceable spares, quarantine/pass bins and shipping.
Circulation/utility allowance	180 sq ft	Access routes, electrical isolation, cleaning and service clearances.

Total planning area: approximately 1,300 sq ft. This is an adjacency/space allowance, not a building or fire-approved layout. Check the actual machines, aisle requirements, lease constraints and local approvals before fit-out. Use one-way case movement from receipt through assessment, work, quality and dispatch.

Power: prepare a simultaneous-load schedule from measured/nameplate printer loads, heaters, extraction, workstations, soldering and test supplies. A licensed electrician sizes circuits, protective devices, earthing, distribution and isolators. Do not size backup power by printer count alone. Base backup should protect IT/job records; whole-farm ride-through is optional after a power study.

Air and cleanliness: printing and finishing need source control, suitable enclosures/extraction and make-up air. Particle filtration and gas/vapor treatment are different requirements. NIOSH recommends appropriate ventilation, containment and electrical/fire controls for small 3D-printing operations. [S13]

Daily discipline: preflight checks, material dryness, bed condition, correct job revision, clear exits, housekeeping, test-zone clearance and end-of-shift shutdown review. Begin with staffed production; validate monitoring and the facility before scheduling unattended work.

Waste: segregate failed prints, electronic scrap, sharps, chemicals and contaminated consumables. Arrange appropriate collection. Resin processing is outside the base layout unless an existing resin cell is separately assessed and equipped.

11 / Factory BOM and procurement

The workbook is the detailed procurement schedule. It includes equipment, finite starter consumables, commissioning allowances and separately gated upgrades. It is complete for the defined launch scope; it cannot be an inventory of every spare for every future object or robot.

Group	What must be covered
Print-cell reliability	Service parts, compatible nozzles/hotends/beds, dryers, sealed storage, racks and printer qualification consumables.
Metrology and CAD capture	Calipers, micrometers, square/height references, gauges, scales, photo setup and calibration. Rent specialist scanning/CMM when needed.
Mechanical fabrication/assembly	Benches, vise, drill/hand tools, torque tools, taps, deburring, controlled finishing, insert installation and test fixtures.
Electronics and robot repair	ESD benches, current-limited supplies, meters, oscilloscope/probes, solder/rework, fume control, microscope, protocol adapters, guarded test area.
Facility and safety	Professional electrical fit-out, source extraction, suitable detection/fire equipment, isolation, PPE and commissioning checks.
Production stock	Qualified filament, fasteners, inserts, bearings, shafts, connectors, wiring, rated modules, adhesives and packaging in defined starting quantities.
Operations and IT	CAD-capable computers net of existing stock, network/backup, labels/barcodes, asset storage and shipping equipment.

How to use the BOM: record usable owned quantity after inspection; calculate net procure quantity; get quotations for the minimum specification; confirm landed cost; approve the needed phase; receive and test; record serial/warranty/calibration where applicable. For lot items, replace the planning lot with a supplier schedule before ordering.

Price treatment: low/high INR ranges are planning estimates, not quotations. Selected retail/manufacture references anchor some lines; tax treatment and scope may differ from the proposed purchase. The workbook identifies estimate versus sourced reference. Exclude GST, inbound freight, duties and premises-specific changes until quoted; add a separate contingency.

Buy in waves: essential measurement, safe workstations and reliability first; replenish materials/components against paid jobs; acquire scanners, advanced printers or production automation only when outsourced spend and lost orders justify them.

Do not buy more printers initially. Audit the ten you own and use the first paid jobs to discover whether measurement, engineering time, a material capability or customer acquisition is the real constraint.

12 / Process and material selection

Route	Good launch applications	Qualification note
PLA / FDM	Visual prototypes, geometry/fit trials and suitable low-demand indoor tools.	Do not assume heat resistance, structural durability or outdoor life.
PETG / FDM	Many enclosures, covers and moderate-duty parts after assessment.	Validate creep, temperature, chemical exposure and layer orientation.
TPU / FDM	Feet, bumpers, flexible covers and vibration interfaces.	Grade/hardness and print behavior matter; not every seal application is suitable.
ASA / ABS / FDM	Suitable outdoor/thermal applications where the exact grade and machine are qualified.	Enclosure and emissions control; shrinkage and fit verification.
Nylon/composites	Selected fixtures and mechanical parts on compatible qualified machines.	Drying, wear-resistant tooling where needed, grade-specific properties and testing.
SLS/MJF partner	More complex nylon shapes or batches unsuitable for available FDM.	Supplier tolerances, finish, dimensional inspection and actual application validation.
CNC / sheet metal partner	Precision interfaces, metal brackets, machined fixtures and sheet housings.	Drawings, material traceability, finishing and inspection defined before order.
Standard bought parts	Bearings, shafts, fasteners, springs, sensors, motors and approved modules.	Prefer rated available components over redesigning established interfaces.

Material approval record: manufacturer/grade, technical data reference, supplier, batch/lot, drying/storage, printer/profile, orientation and application restrictions. Avoid a universal statement that "nylon" or "carbon fiber" is strong enough; the complete part and process determine performance.

DFM review: minimum walls, unsupported geometry, insert retention, screw direction, layer loading, sharp corners, clearances, distortion, accessible inspection points and post-processing. If the original part failed, understand the failure before copying its geometry.

PCB route: IoT kits support prototyping. For a new customer board, define requirements, schematic, layout, design checks, fabrication/assembly files, BOM and bring-up procedure. Outsource PCB fabrication/assembly; charge engineering separately. Avoid launching a general PCB factory.

Food/medical requests: a material label alone does not validate final-part hygiene, cleanability or biocompatibility. Application-specific validation and suitable specialist manufacturing are needed; these are outside the launch offer. [S14]

13 / Manufacturing and quality SOP

Gate	Evidence	Owner
1. Accept scope	Use, failure consequence, source/make comparison and capability route.	Intake engineer
2. Freeze specification	Units, datums, critical dimensions, load/environment and pass/fail criteria.	Engineer
3. Release design	CAD/drawing revision, material/process selection and DFM review.	Engineering lead
4. Approve commercial work	Exact quote and customer-facing design/scope revision accepted.	Customer/case owner
5. Release production	Qualified machine/material/profile, inspection plan and approved quantity.	Factory lead
6. First article	Measured critical dimensions and representative fit/function test.	QC/engineer
7. Dispatch	Complete traveler, final checks, instructions, labels and release identity.	QC/release lead

Work traveler: case and part IDs; design revision; quantity; material lot; machine ID; profile/slicer version; build orientation; G-code hash; operator; start/finish; actual hours/material; post-process; inspection results; nonconformances and release.

First fit: inspect against the mating component where possible. If only customer measurements are available, identify that limitation and use a paid fit prototype before a batch. Record customer-installed acceptance separately from the workshop dimensional pass.

Batch checks: establish sampling or 100% inspection based on critical features, lot size and consequence of failure. During the pilot, inspect every critical interface on small batches. Keep a retained reference part where useful. A failed first article blocks the remaining batch.

Nonconformance: quarantine the part, record defect/cause, decide rework/reprint/referral, communicate schedule impact, and re-inspect. Do not silently widen a tolerance after printing. Material, geometry or process changes follow revision control.

Warranty proposal: define a limited workmanship/approved-fit remedy period, such as 30 days for the pilot, with exclusions and remedy process clearly drafted for review. Preserve applicable consumer rights; do not imply a repaired entire machine receives an unrestricted warranty. Every claim gets a case, failure evidence and engineering review.

Dispatch pack: part/asset ID, approved-use statement, install/use instructions, test summary, safe limitations, reorder link and support contact. Track packaging damage separately from design or manufacturing failure.

14 / Robot and electronics repair SOP

Supported-family list: begin with the lab/education/IoT equipment the team can document and safely test. Add a family only when manuals, diagnostic access, spare supply, trained technicians, test fixtures and final acceptance criteria are available.

- **Receive:** photograph condition, model/serial, accessories and visible damage; record customer symptom, prior work and data/firmware consent. Keep incoming assets separate until inspected.
- **Make ready:** isolate energy, inspect cabling and batteries, identify unexpected motion/power hazards, and establish a controlled test setup. Route damaged/high-energy batteries to an appropriate specialist.
- **Diagnose:** reproduce the fault where safe; distinguish mechanical, power, wiring, sensor, controller and software/configuration causes; record measurements and confidence. A broken cover may not explain an intermittent sensor.
- **Estimate:** issue findings, required parts/labor, uncertainty, optional work, turnaround and no-fix outcome. Obtain approval before work outside paid diagnostic scope.
- **Repair:** replace a supported module/harness or perform competent low-voltage board work; manufacture approved mechanical parts; retain removed parts as agreed. Record serials and configuration changes.
- **Verify:** restore approved configuration, test every repaired function, execute a defined repeated task/cycle test and inspect thermal/current/mechanical behavior against agreed limits.
- **Return:** give fault/action/test report and maintenance guidance. Arrange asset-specific recommissioning by a qualified person when installation affects operational safety.

Example test plan: for a small lab robot with a broken sensor mount and intermittent connector, verify mount interfaces, cable retention, sensor readings, movement clearance and repeated operation under the agreed workload. Define the cycle count and acceptance threshold for that job; a generic "burn-in passed" label is insufficient.

Bench practice: use ESD controls, suitable meters/probes, current-limited supplies and soldering extraction. Never defeat oscilloscope protective earth to make a floating measurement; use appropriate measurement equipment and trained procedures. [S15]

Industrial robotics pathway: first supply noncritical peripherals with integrator review; then establish an OEM/specialist service relationship. Servo drives, joint calibration, brakes and safety functions remain under specialist procedures with post-repair validation. [S9]

15 / Staffing and accountability

Role	Monthly planning cash	Primary accountability
Mechanical/CAD lead	INR 75,000	Scope acceptance, CAD, DFM, material selection, inspection criteria and design release.
Mechatronics/repair engineer	INR 55,000	Supported robot diagnosis, electronics, test rigs and functional validation.
Print/assembly technician	INR 30,000	Printer upkeep, job execution, materials, assembly and first inspection.
Customer/operations coordinator	INR 30,000	Intake completeness, quotes, updates, purchasing, logistics and records.
Founder	INR 0 cash draw in base model	Sales, partnerships, product ownership, escalation and financial discipline. Editable assumption.
Software team	Existing team allocation assumed	Portal MVP and integrations; effort has an opportunity cost even when no new cash salary is added.

Payroll budget: INR 1.90 lakh/month in the base model. These are planning allowances, not local salary quotes. Add employer on-costs, benefits, recruiting and founder salary where applicable before approving a budget. A smaller pilot can combine roles but must state the available hours and competence.

Engineering capacity: two engineers x 160 paid hours/month x 65% customer-job availability = 208 hours. The example assumes the repair engineer is qualified to contribute 24 hours/month to CAD/fixture work: mechanical capacity becomes 128 hours and repair capacity 80. This is a hiring/skills assumption to validate. Without it, the mechanical lead alone has 104 hours and cannot deliver the example mix.

Operating rhythm: daily 15-minute case/production review; weekly contribution and rework review; weekly material/maintenance planning; fortnightly customer interviews; monthly calibration/qualification status review. Record blocked work with a reason and next owner.

Supplier qualification: select at least one practical source plus a backup for critical consumables and outsourced processes. Verify capability with a test order, traceability, inspection results, lead time, confidentiality, change control and nonconformance response.

Before opening: obtain premises-specific electrical/fire/extraction advice; confirm applicable business, tax, waste and employment obligations; review customer terms, IP permissions, insurance, asset custody and warranty handling with local professionals. These are launch tasks, not a claim that one universal license list applies.

Capability record: maintain a technician skills matrix, authorized procedures, supported equipment families and external specialist contacts. Add services only when someone is explicitly responsible for competence and release.

16 / Pricing and unit economics

Offer	Illustrative starting band, ex tax/shipping
Simple assessment / remote triage	Free limited screening; no promise of design work.
Bench diagnosis	INR 750-2,500 for a defined initial task; complex work quoted separately.
Simple reverse engineering/design	INR 1,500-5,000; complex interfaces and multiple iterations individually scoped.
Simple replacement order	INR 1,500-4,000 typical planning basket including appropriate design allocation.
B2B jig/fixture job	INR 6,000-25,000 depending design, quantity, material and validation.
Supported robot repair	INR 3,000-20,000+ depending diagnosis, parts, labor and tests.
Retainer, after repeat demand	Defined hours/response window and billing minimum; production and spares priced separately.

These are price-testing hypotheses, not market quotations. A small clip can be cheap to print and expensive to measure/design. State what the engineering fee includes: deliverable, revisions, prototype quantity and acceptance procedure. Rework caused by Tobor error differs from a customer-requested change.

Quotation floor: material and expected scrap + purchased/outside services + packing/shipping + transaction fees + engineering/operator time allocation + machine wear allocation + complexity/rework allowance. Compare the result with customer value and alternatives. Use an explicit minimum order/design charge.

Two views of margin: cash contribution is revenue less incremental job cash costs. Economic job margin also allocates engineering and machine time to test whether the work uses capacity well. Regular payroll is already in monthly overhead; do not subtract allocated payroll twice when calculating business cash surplus.

Illustrative simple order: INR 2,500 revenue less INR 900 incremental costs = INR 1,600 contribution. At 0.6 engineering hours x INR 750/hour, the engineering allocation is INR 450; remaining contribution after that allocation is INR 1,150 before machine/operator allocation and other overhead. The workbook lets you edit all inputs.

Cash terms: take payment for diagnosis/design before work; use a production deposit for custom or bought-in parts; collect balance before dispatch where agreed. Any partial credit of diagnosis must appear explicitly in the quote. Price partner/referral work transparently, including who carries warranty responsibility.

17 / Operating model and capacity

Monthly job mix	Revenue	Variable cost	Contribution
100 replacements x INR 2,500	250,000	90,000	160,000
20 B2B jigs x INR 12,000	240,000	70,000	170,000
15 robot repairs x INR 10,000	150,000	60,000	90,000
25 repeats x INR 3,000	75,000	22,500	52,500
Total, 160 orders	715,000	242,500	472,500

Base recurring fixed cash cost: payroll INR 190,000; rent INR 45,000; utilities/filter provision INR 20,000; software/cloud INR 18,000; marketing/general logistics overhead INR 22,000; professional/insurance/admin/calibration provision INR 15,000. Total INR 310,000/month. These grouped allowances need quotations. Job freight and outside processes belong in variable costs; do not duplicate per-job consumables as overhead.

Illustrative operating cash surplus: INR 472,500 - INR 310,000 = INR 162,500/month before founder draw, tax, interest, depreciation and any unbudgeted employer/software costs. At this unchanged mix, cash break-even revenue is approximately INR 4.69 lakh/month. This depends on actual contribution and capacity; it is not guaranteed profitability.

Printer planning: 10 printers x 26 days x 16 scheduled hours x 80% availability x 90% first-pass yield x 50% commercial utilization = 1,497.6 good sold print-hours/month. Sixteen scheduled hours requires an approved staffing/monitoring arrangement. Use a smaller schedule initially if that is not available.

Engineering bottleneck: the mix uses 123.75 CAD/fixture hours and 45 repair hours. It is feasible only with qualified cross-support: transfer 24 repair-engineer hours to CAD, giving 128 CAD and 80 repair hours. CAD then has just 4.25 hours spare. At unchanged mix the discipline limit is about 1.03x, despite 39.25 hours of pooled headroom. Without cross-support, the base mix exceeds CAD capacity.

Sensitivity: losing INR 500 contribution on every order would reduce monthly surplus by INR 80,000. A founder draw of INR 100,000 would reduce the base surplus to INR 62,500. More printer utilization will not resolve an engineering or demand constraint.

Expansion rule: add the resource causing measured lost contribution. A machine purchase needs confirmed compatible work, full operating/inspection costs and expected utilization; a recurring engineer backlog may justify staffing or template work first.

18 / Cash and investment gates

Budget structure: the workbook separates launch setup, starter stock, optional expansion, recurring overhead, refundable/working-capital items and portal development. Every owned usable tool reduces procurement. The detailed item schedule is the authoritative planning basis; no vendor quotation has been obtained.

Component	Low INR lakh	High INR lakh
Launch equipment + setup + opening stock	7.57	13.70
15% planning contingency	1.14	2.06
Setup including contingency	8.70	15.76
Refundable premises deposit assumption	1.35	1.35
Three-month fixed operating cash reserve	9.30	9.30
Additional variable-cost working float	1.00	2.00
Planning cash envelope (base software assumption)	20.35	28.41
Optional external portal build: additional	6.00	6.00

Runway method: reserve three months of the base fixed cash costs (INR 9.30 lakh), plus a defined variable-cost float. This is a conservative cash buffer, not an additional expense on top of three months of modeled operating losses. The workbook also shows an illustrative ramp so the two views can be compared without double counting.

Software choice: allocate an existing team to the concierge/MVP scope if capacity exists. The base cash model assumes no new upfront cash fee; it does not imply free effort. Use a separate planning allowance of INR 6 lakh for an external MVP implementation if needed, subject to a scoped quote. Avoid commissioning a full marketplace before paid demand.

Gate A - pilot essentials: after inventory and premises review, purchase only the measurement, safe workbench and reliability gaps required for the first supported jobs. Use existing tools and partners for low-frequency capabilities.

Gate B - operating unit: after first paid jobs and supplier checks, complete the required print-cell, electronics, inspection, storage and logistics setup. Procurement remains tied to actual condition and service mix.

Gate C - expansion: scanners, advanced material printers, resin equipment, heavy backup power and higher-volume automation require a separate capability case. Professional CNC, SLS/MJF, metal fabrication and specialist robot work remain outsourced initially.

Not yet included in an approved funding number: actual taxes/credits, inbound freight and duties, precise premises modifications, legal/local approval fees beyond allowances, extended hiring costs, owner salary, larger robot handling/transport and any required existing-printer replacement. Resolve these from quotations and the audit before committing capital.

19 / The first 90 days

Period	Work and owner	Evidence / deliverable
Days 1-14	Founder + leads: inventory, premises/capability check, 30 interviews across factories, labs and repair shops; collect real jobs.	Inventory and capability matrix; 15 candidate jobs with price/lead-time context; chosen supported scope; 6 demonstration jobs.
Days 15-30	Engineers + ops: simple intake page, manual assessment/quote/job cards, baseline qualification, essential procurement.	Target 10 paid jobs from 5 unrelated buyers; actual hours/costs/fit results; stop/change decision on poor-fit niches.
Days 31-45	Product/software + ops: cases/assets, quotes, approvals, file versions, work queue and payments.	End-to-end rehearsal from submission through refund/rework/dispatch; no orphan approvals or unreconciled payments.
Days 46-60	Factory + software: QC records, private part library/reorder, cost capture and partner process; standardize two job families.	Target 25 cumulative paid jobs and 3 repeat buyers; verified first-article records and contribution evidence.
Days 61-90	Founder + team: deepen winning segments, improve guided measurement/template CAD, measure bottlenecks and adjust pricing.	Target 40-60 cumulative paid jobs; launch decision based on fit, delivery, contribution and repeat purchase.

Week-one start pack: printer/tool inventory; two-segment offer sheet; risk/acceptance matrix; intake checklist; source-versus-make quote template; job traveler; inspection sheet; first six demo briefs; supplier shortlist; and a named owner for each workflow.

Dependencies: equipment qualification before paid manufacturing release; paid diagnosis before final repair estimate; measured requirements before approved CAD; approved CAD/quote before production; inspection before dispatch; repeated successful jobs before automatic reordering.

Capacity during the pilot: the 40-60 cumulative jobs target is learning volume, not the 160-order illustrative steady month. Keep enough engineering time for qualification, documentation and customer feedback. Do not use the steady-state economics as a month-one sales promise.

Critical path: qualified mechanical/repair leads, reliable input measurements, material/process capability, customer willingness to pay engineering fees, and physical fit validation. The website is only one workflow.

Execution starts with an inventory audit and paid concierge jobs. The plan does not require waiting for sophisticated AI CAD or a complete marketplace.

20 / Go-to-market and decision gates

Sales motion: founder-led visits to reachable factory/automation teams; repair-shop partnerships that consolidate orders; lab/robotics demonstrations; specific search pages for supported parts and use cases. Publish measured case studies with permission. Do not spend heavily on broad consumer advertising before knowing design and delivery costs.

Metric	Proposed day-90 target	Definition
Paid pilot volume	40-60 cumulative jobs	Completed paid work; show cancellations/refunds separately.
Repeat demand	At least 3 unrelated repeat buyers	A second paid order, not a promise or an unused account.
First-fit acceptance	At least 85%	Parts accepted on first installation attempt / parts with recorded installation outcome; report unknown outcomes separately.
On-time delivery	At least 90%	Jobs delivered by the confirmed date / jobs due; log approved date changes separately.
Quote conversion	At least 25%	Accepted quotes / qualified quoted requests, segmented by service.
Positive job economics	Per chosen service family	Include actual design/repair time allocation, outside costs, scrap, packing, shipping and rework.
Engineering productivity	Improves on repeat families	Contribution per engineering hour and reduced redesign time; no fixed AI-speed claim.

Pause or change the offer if: customers consistently reject design fees; custom consumer work costs more than replacement; hidden dimensions cause repeated reprints; unsupported OEM access blocks robot jobs; or demand is concentrated in work outside safe capability. Change niche, intake, pricing or partners before adding machinery.

Month 4-6: formalize limited business retainers for customers with observed repeat needs; build 50-100 verified part records only as paid work generates them; add supply alternatives and template families; evaluate one capability upgrade based on lost/outsourced work.

Month 7-12: consider one additional city/partner cell after stable first-fit, delivery and margin results. Explore OEM-authorized spares, repair-shop white-label service and fleet asset accounts. A partner needs the same job package, inspection evidence and service responsibility as the owned workshop.

Longer term: offer secure digital spare libraries, reorder APIs for factories, asset maintenance histories and local production under approved processes. Predictive maintenance needs real fleet failure data and customer permissions; do not make it an initial sales claim.

Decision now: review this plan and choose the first two service families. During execution, confirm printer/tool inventory, workshop city/space, available engineers/software capacity and spending ceiling. Those facts turn the BOM ranges into a purchase-ready specification.

21 / Sources and planning notes

Sources support the specific external capabilities and technical considerations cited in this plan. Budgets, pricing, staffing, sales targets, layout and business recommendations are planning assumptions and proposals. Vendor claims are not independently measured service benchmarks. The workbook contains its own item-level price reference register.

- [S1] UltiMaker: application focus, jigs and fixtures
- [S2] Stratasys: jigs and fixtures application support
- [S3] Fracktal Works: engineering/manufacturing services
- [S4] 3Ding: services
- [S5] Xometry: design services
- [S6] ABB: robot repair services
- [S7] Philips: official Printables profile / Fixables
- [S8] Government of India: Right to Repair Portal, July 2024
- [S9] Universal Robots: maintenance and repair manual
- [S10] CadQuery: documentation and CAD exports
- [S11] EINSTAR: 3D scanning vs photogrammetry
- [S12] Prusa: modeling with 3D printing in mind
- [S13] NIOSH: approaches to safe 3D printing, small businesses
- [S14] Formlabs: food-safe and biocompatible material restrictions
- [S15] Tektronix: floating oscilloscope measurements and protection
- [S16] Autodesk: Fusion personal-use terms

Commercial CAD licensing must be checked for the chosen tool. Autodesk Fusion personal use is intended for qualifying noncommercial use and should not be assumed available for this commercial workshop.
[S16]

Financial figures are rounded in prose. The editable workbook formulas govern the illustrated calculations. Obtain current written quotations with specifications, taxes, delivery and warranty before ordering.